EE3032 Introduction to VLSI Design—Homework 3

1. Design an 8-to-1 multiplexer using transmission gates driving a capacitive load 10fF, where the transmission gate is implemented with 10μm/0.18μm NMOS and 20μm/0.18μm PMOS transistors. Simulate your design using Hspice simulation and measure the delay of the multiplexer. Then, modify the design such that the delay of the new design is smaller than the original design without changing the sizes of the transmission gate. (5 points)

2. The dynamic power consumption (P_d) of a CMOS circuit is composed of switching power consumption P_{sw} and short-circuit power consumption (P_{sc}). Consider an inverter driving a load C_L shown in Fig. 1. Complete the curve of P_{sc} associated with C_L, and the value of C_L is ranged from 0.1pf, 0.3pf, 0.5pdf, 0.7pf to 1pf using Hspice. Check if the P_{sc} is associated with the C_L. If yes, why? (5 points)

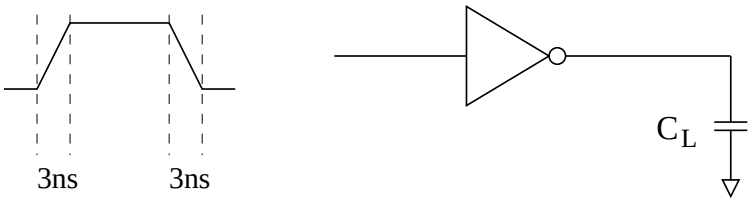

Figure 1: Inverter.